1A) three directions

2A) VLSI testing, traffic control, parallel memory storage schemes and deadlock prevention.

3A) Hamiltonian path problem is a problem of finding a path in a graph that visits every node exactly once.

4A) a) create a leaf node for each unique character along with and build a min heap of all the leaf nodes. The value of frequency field used to compare two nodes in min heap.

b) Extract 2 nodes from min heap with min frequency.

c) Create new internal node with frequency equal to sum of two nodes.

d) Repeat step # b and step # c until heap contains only one node. The remaining node is root and tree is complete.

5A) O(nlogn) is the time complexity of tree vertex splitting problem.

II.

1A) The clique problem is the computational problem of finding cliques (subsets of vertices, all adjacent to each other, also called complete subgraphs) in a graph.

→ It has several different formulations depending on which cliques and what information about cliques should be found.

→ A clique is a subgraph of graph such that all vertices in subgroup are completely connected with each other.

2A) NP complete problems have below two properties

i. Any given solution to the problem can be verified quickly.

ii. If the problem is solved quickly, then every problem in NP can be solved quickly.

3A) Examples of polynomial time:

$O(n^2), O(n^3), O(1), O(n \log n)$

4A) 1. Initialize the result as {}

2. Consider a set of all edges in given graph. Let the set be E.

3. Do the following while E is not empty.

a) Pick an arbitary edge (u,v) from set E and u,v to result

b) Remove all edges from E which are incident on u or v.

4. Return result

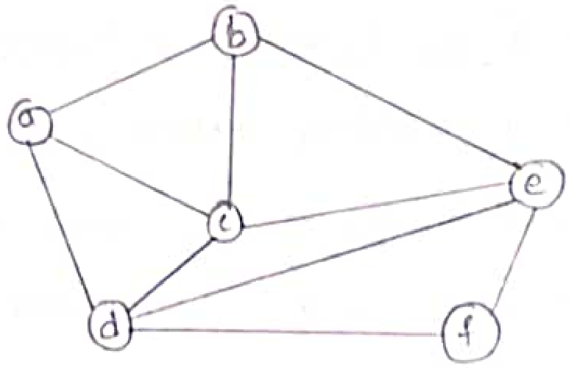
5A) a) Decision problem - In this we search for a feasible solution

b) optimization problem - In this we search for best solution.

c) Enumeration problem - In this we find all feasible solutions.

III

1A) Given a graph G=(V,E) we have to find the Hamiltonian circuit using Back tracking approach we start our search from any arbitary vertex say 'a'. This vertex 'a' becomes the root of our implict tree. The first element of our partial solution is first intermediate vertex of the Hamiltion cycle to be constructed. The next adjacent vertex is selected by alphabetic order.

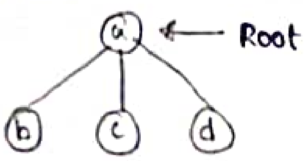


If at any stage arbitary vertex makes a cycle with any vertex other than vertex 'a' then we say dead end is reached. In this case, we backtrack one step and again search begins by selecting another vertex. The search using backtracking is successful if a Hamiltion cycle is obtained.

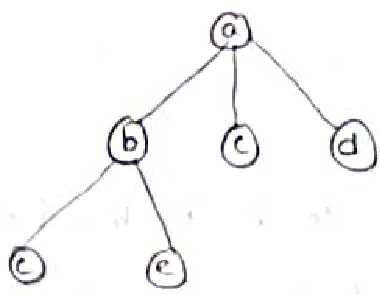
Example: consider a graph $G = (V, E)$ we have to find Hamiltion circuit using Backtracking method.

Solution: First we start our search with vertex 'a' and there vertex becomes root of our implict tree.

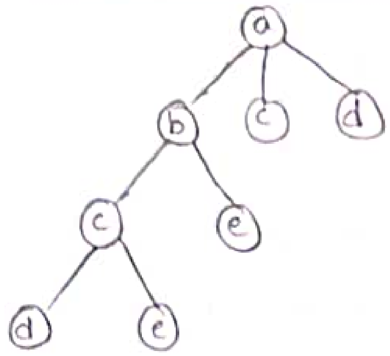
Next, we choose vertex b adjacent to a as it becomes first in lexlographical order (b, c, d)



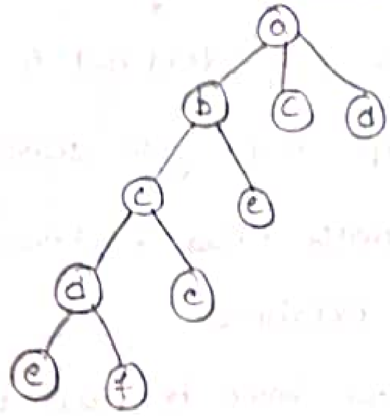
Next, we select 'c' adjacent to b:



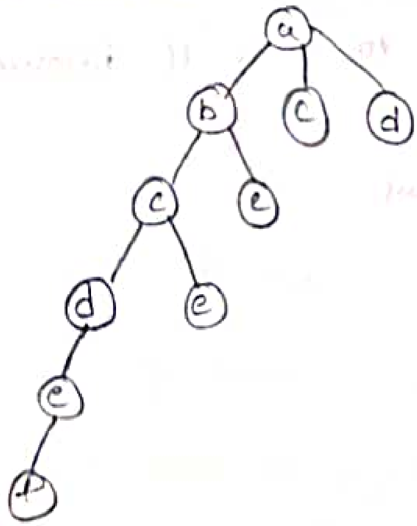
Next we select 'd' adjacent to 'c'



Next we select 'e' adjacent to 'd'.

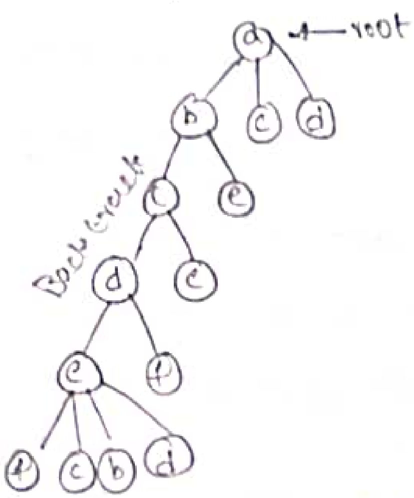


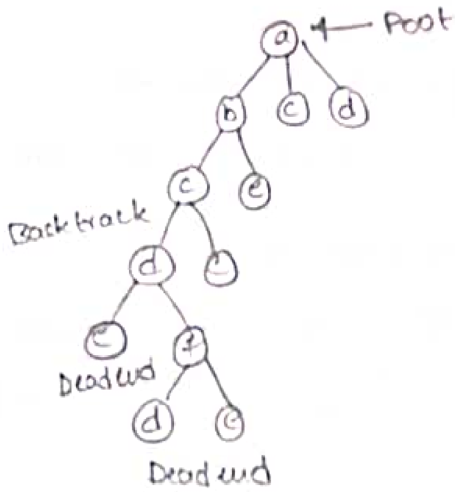
Next we select 'f' adjacent to 'e'. The vertex adjacent to 'f' is d and e but they have already visited. Thus we get the dead node end and we backtrack one step and remove vertex 'f' from partial solution.



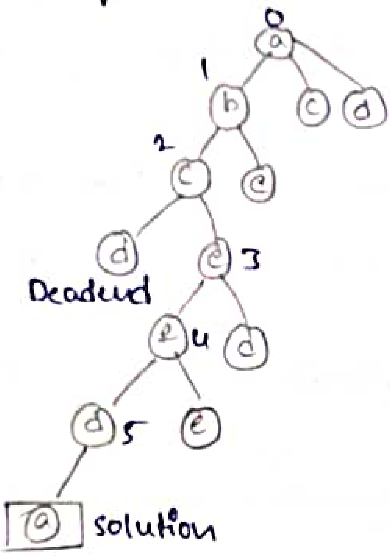
Dead end

from back tracking vertex adjacent to 'e' is b,c,d and f from which vertex 'f' has already checked and b,c,d already visited. So, again Backtrack one step. Now vertex 'f' adjacent to d are e. frome has already checked, the adjacent of 'f' are d and e frame has already checked. If 'e' vertex revisited we get dead state. Here we get the Hamiltion cycle as all vertex other than start vertex 'a' is visited only one (a-b-c-e-f-d-a).





Again Back track



Here we have generated one Hamilton circuit, but another Hamilton circuit also be obtained by considering another vertex